

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech. (IT) Examination****June 2022****IT256 Data Structures & Algorithms****Date: 22.06.2022, Wednesday****Time: 10:00 a.m. To 1:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q - 1 Do as directed.**

- 1) True or False: For linear search, data must be in ascending order. [01]
- 2) Define: Input Restricted Deque. [01]
- 3) Fill in the blank. A variant of a linked list in which the last node of the list is having the address of first node of the list is _____. [01]
- 4) What are the differences between array and linked list? [02]
- 5) What is Data Structure? Discuss different types of Data Structure with example. [03]

Q -2(a) For a given 3-D array A[-2:2, 1:4, 6:9] with base address 2000 and size of element is 4 bytes. Find out total number of elements and address of A[1, 3, 8] element in column major order. [05]

- (b)** (i) Explain Priority Queue with suitable example. [04]
(ii) Consider the following operation performed on a stack of size 5. [03]
Push(1), Pop(), Push(2), Push(3), Pop(), Push(4), Pop(), Pop(), Push(5), Push(3)
Draw the Stack after each operation. After the completion of all operation, How many elements are present in stack?

OR

- (b)** (i) Evaluate the following expressions using stack. [05]

$$3 \ 2 + 9 \ 4 / 2 / 6 \ 2 * 3 + * -$$

- (ii) Draw Doubly Linked List with four nodes. List out two applications of Doubly Linked List. [02]

Q - 3 Answer following questions. (Any Three)

[15]

- 1) Explain Tower of Hanoi problem using stack for $n=3$ disk.
- 2) Consider a Singly Linked List of five integer elements. Write an algorithm/pseudocode to insert at front and insert at end operation for the given Singly Linked List.
- 3) What is Hashing? Explain one collision resolution technique with suitable example.
- 4) What is Circular Queue? Discuss overflow and underflow conditions of Circular Queue.

SECTION - II

Q - 4 Do as directed.

[01]

- 1) Fill in the blank.

Stack data structure is used in _____ traversal technique of graphs.

- 2) How many comparisons are required for n items in selection sort? [01]
- 3) True or False: Every complete binary tree is also a full binary tree. [01]
- 4) Define : Connected Graph [01]
- 5) In a Binary Search Tree, preorder traversal is 50, 20, 30, 25, 65, 55. What is the in-order traversal of the tree? [01]

Q -5(a) Construct Binary Search Tree for the given data. Apply preorder and post-order traversals on the generated Binary Search Tree. [05]

45, 23, 29, 85, 92, 7, 11, 35, 49, 51

(b) Answer following questions (Any Two)

[10]

- 1) Create Binary tree from following order.

Inorder : G, D, H, L, B, E, A, C, I, F, K, J

Preorder : A, B, D, G, H, L, E, C, F, I, J, K

- 2) Apply binary search on below given data for searching element 9.

9, 11, 20, 26, 28, 32, 34, 44

- 3) Apply bubble sort and show the output after every pass.

42, 35, 12, 77, 5, 101

Q - 6 Answer following questions (Any Three).

[15]

1) Trace the following numbers using radix sort.

9, 1106, 9099, 6, 7101, 990, 15, 99, 10, 909

2) Define graph. Explain storage representation of a graph.

3) Create AVL Tree for the given numbers. Show the updated tree after each operation.

7, 1, 6, 2, 5, 4, 3

4) What is Max Heap Tree? Construct Max Heap Tree for following data.

103, 11, 101, 110, 111, 119, 19, 91
